

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

#XUV♥

CHEMICAL BONDING AND MOLECULAR STRUCTURE

LECTURE-4 CHEMISTRY

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Topics To Covered

1 Bond parameters

2 Resonance

★ (Imp).

REVISION



Drawing Lewis dot Structure $\rightarrow$

Rules $\rightarrow$

$$1) \text{ TVE} = \text{Valence } e^- + (\text{+ve charge}) - (\text{+ve charge})$$

of each atom

2) Central atom $\rightarrow$

$\downarrow$
Least no.

$\downarrow$
Least EN-element
(H & F central atom X)

$$3) \text{ Total no. of bonds} \rightarrow \frac{\text{Reqd. } e^- - \text{TVE}}{2}$$

Formal charge calculation

$$FC = V - \frac{L}{2} - L$$

$\downarrow$ $\downarrow$

Limitations of Octet Rule $\rightarrow$

- Hydrogen
- Hypovalent species.
- Hypervalent species.
- Odd e^- species.
 NO, NO_2
- Compounds of Noble gas.

Bond Parameters: →

a) Bond length → It is distance between two nuclei of bonded atoms in a molecule. It is measured in terms of picometer or Å. with the help of X-ray spectroscopy.

Factors affecting bond-length →

i) Size of the atoms → As size of the atoms increases, bond length also increases. e.g.

$$\text{H-F} < \text{H-Cl} < \text{H-Br} < \text{H-I}$$

ii) Bond Multiplicity → As no. of bonds between two atoms increases, Bond length decreases because of increase of force of attraction.

e.g. $\text{C}-\text{C} > \text{C}=\text{C} > \text{C}\equiv\text{C}$
 (Bond multiplicity $\propto \frac{1}{\text{BL}}$)

b) BOND ENERGY or BOND ENTHALPY: →

To understand bond energy, we have to discuss bond dissociation energy first. Bond dissociation energy → Energy required to break a one mole of one type of a bond is called bond dissociation energy.

BOND ENTHALPY → It is the average of Bond energies of all the similar kind of bonds present in a molecule.

In case of diatomic molecule bond dissociation energy is equal to the bond enthalpy.

e.g. $H_2, Cl_2, F_2, Br_2, O_2$ etc.

In case of polyatomic molecules e.g. CH_4 .

$$\text{Bond Enthalpy} = \frac{BDE_1 + BDE_2 + BDE_3 + BDE_4}{4}$$



Unit = KJ/mol

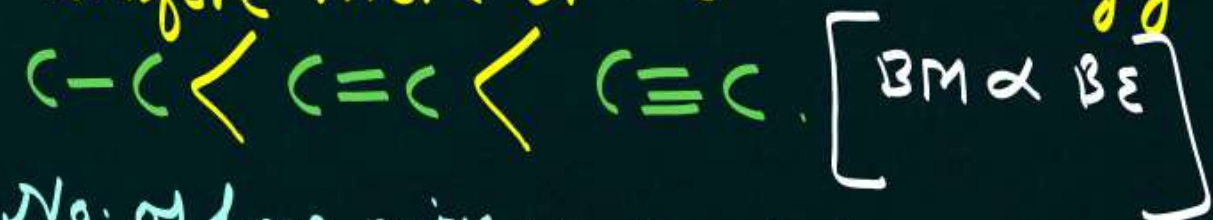
Factors affecting Bond enthalpy $\rightarrow$

i) Bond length $\rightarrow$ Shorter is the bond length, more is the force of attraction between two atoms, more is bond enthalpy. e.g.



$$(BL \propto \frac{1}{BE})$$

ii) BOND MULTIPLICITY $\rightarrow$ More the bond multiplicity between two atoms, more is the force of attraction, therefore more is the bond energy. e.g.



iii) No. of lone pairs $\rightarrow$ Due to the presence of lone pair there is repulsion between two bonded atoms, therefore, lesser is the bond energy e.g.



c) Bond Angle $\rightarrow$ It is angle present between two bond pairs of electrons in a molecule.

It is denoted by θ .

Bond angle is effected by presence of lone pairs (More the lp, lesser is the θ)
e.g. CH_4 , NH_3 , H_2O .



d) Bond Order $\rightarrow$ It is no. of bonds present between two atoms. e.g.

H_2 ($\text{H}-\text{H}$), Bond Order = 1.

O_2 ($\text{O}=\text{O}$), Bond Order = 2.

N_2 ($\text{N}\equiv\text{N}$), Bond order = 3.

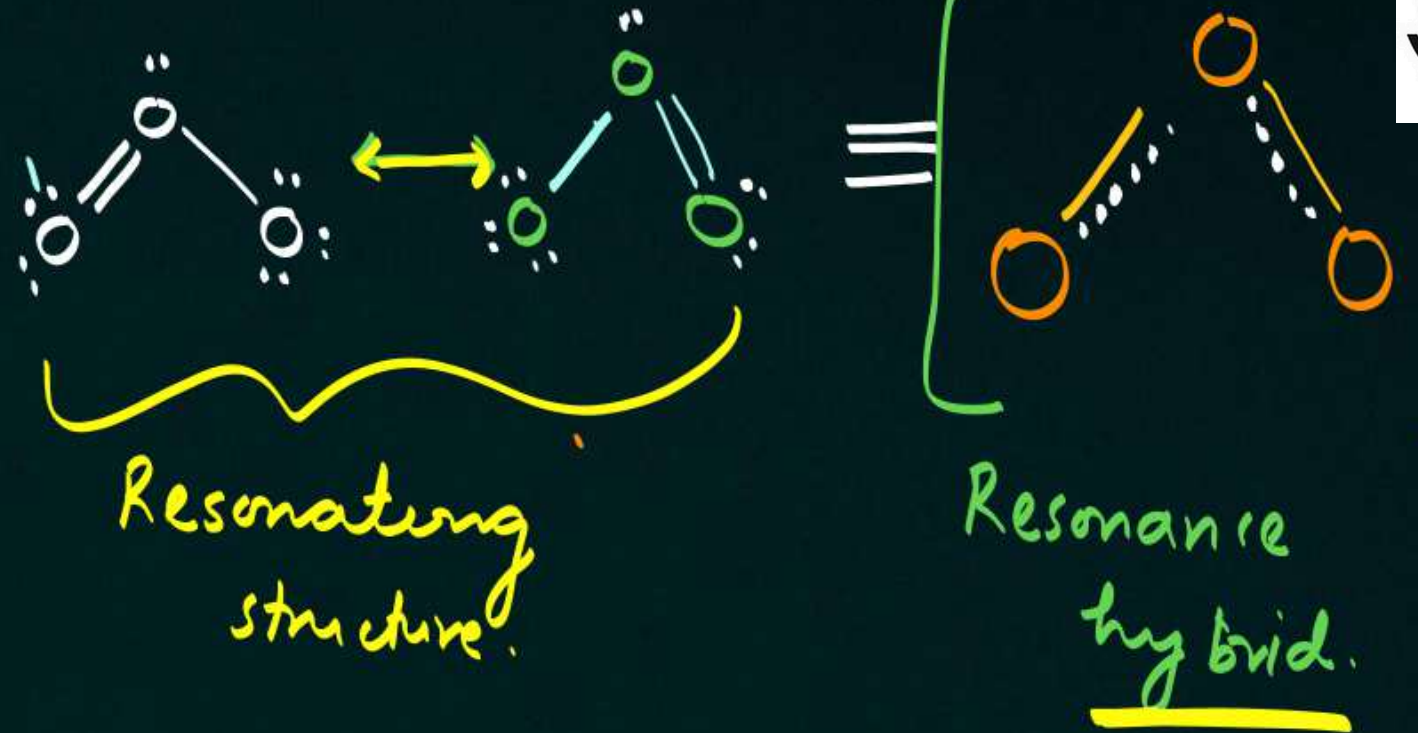
★★

RESONANCE

When one Lewis dot structure of an molecule cannot explain all of their properties, then we draw its more than structure called resonating structures (canonical forms).

The real structure is the hybrid of all the resonating structures is called resonance hybrid. The whole phenomena is called. RESONANCE.

For example in case of ozone.



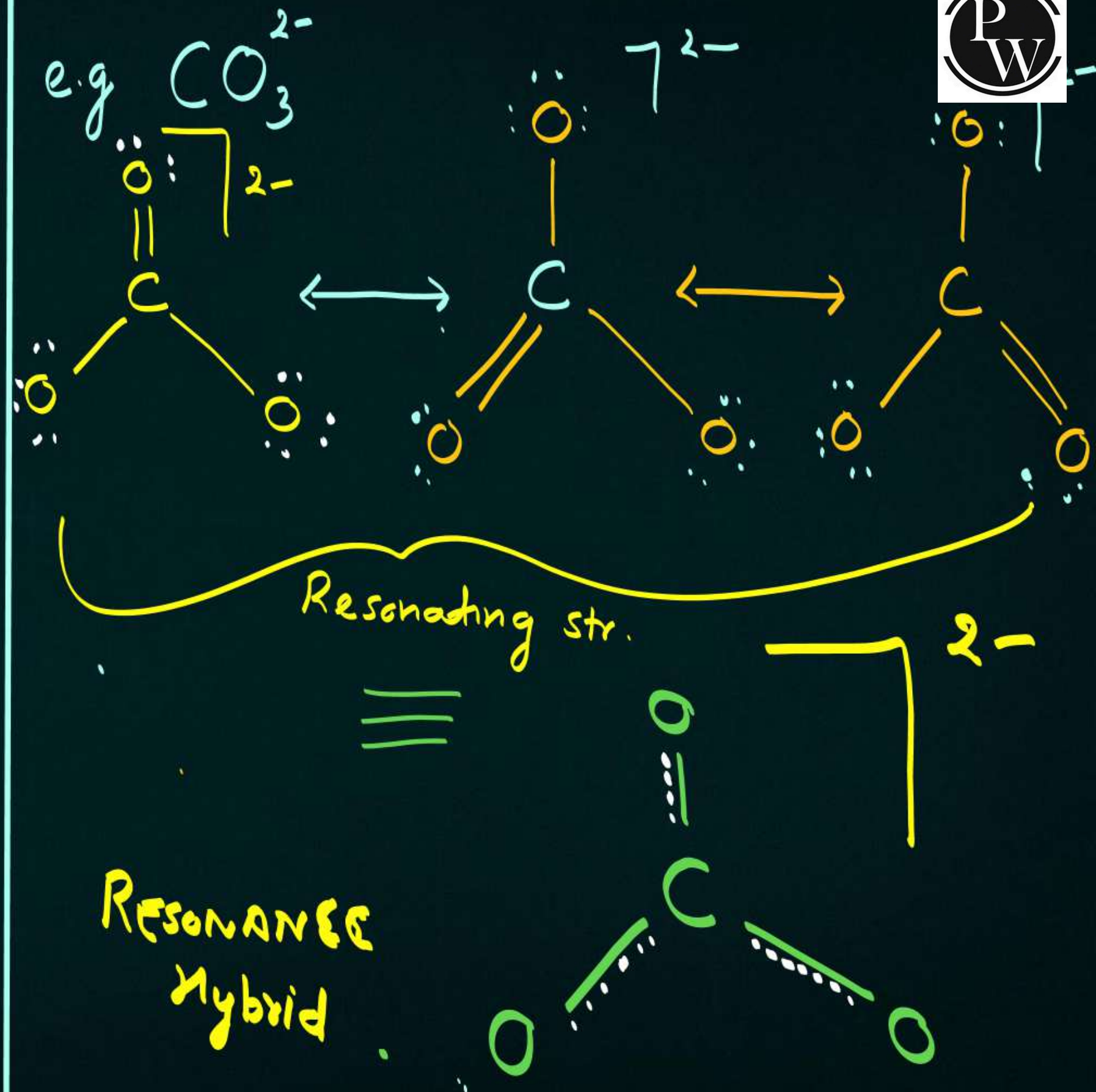
It must be noted that

- Resonance hybrid is real structure, not the resonating structure.

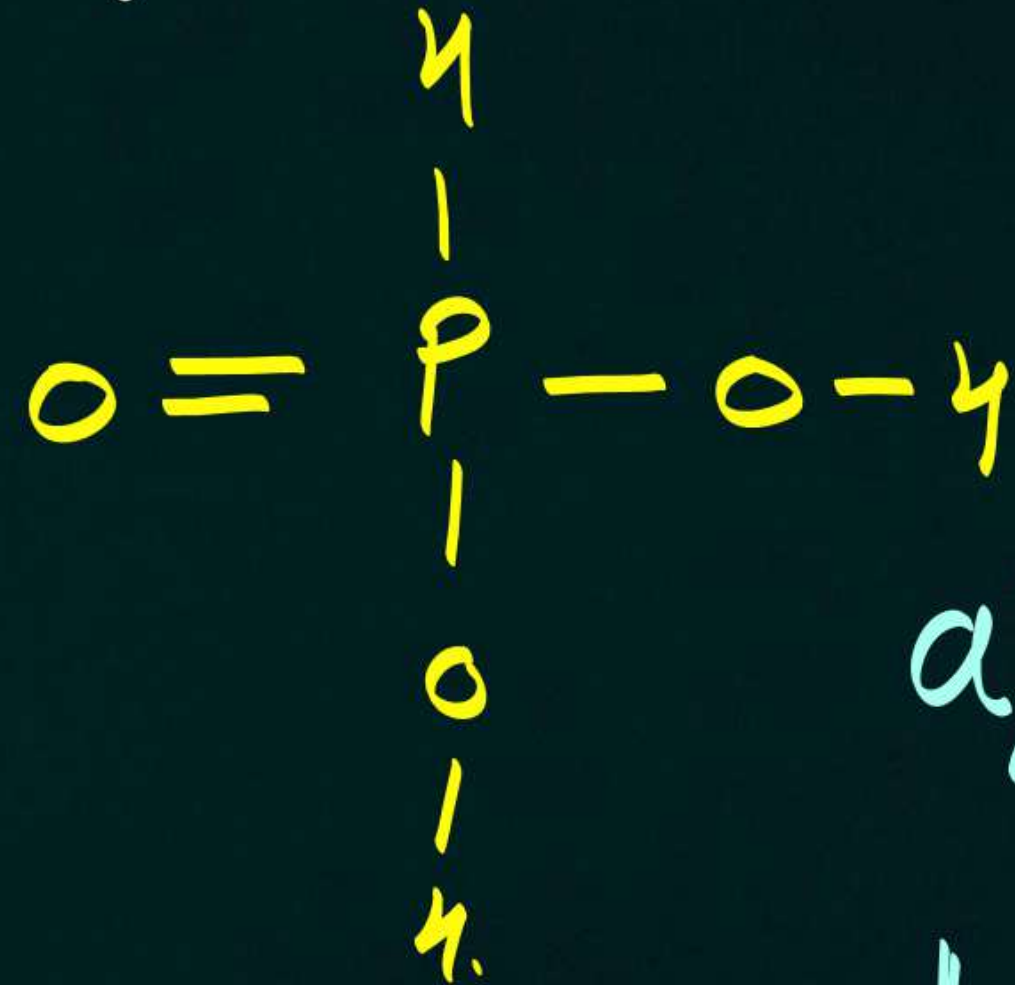
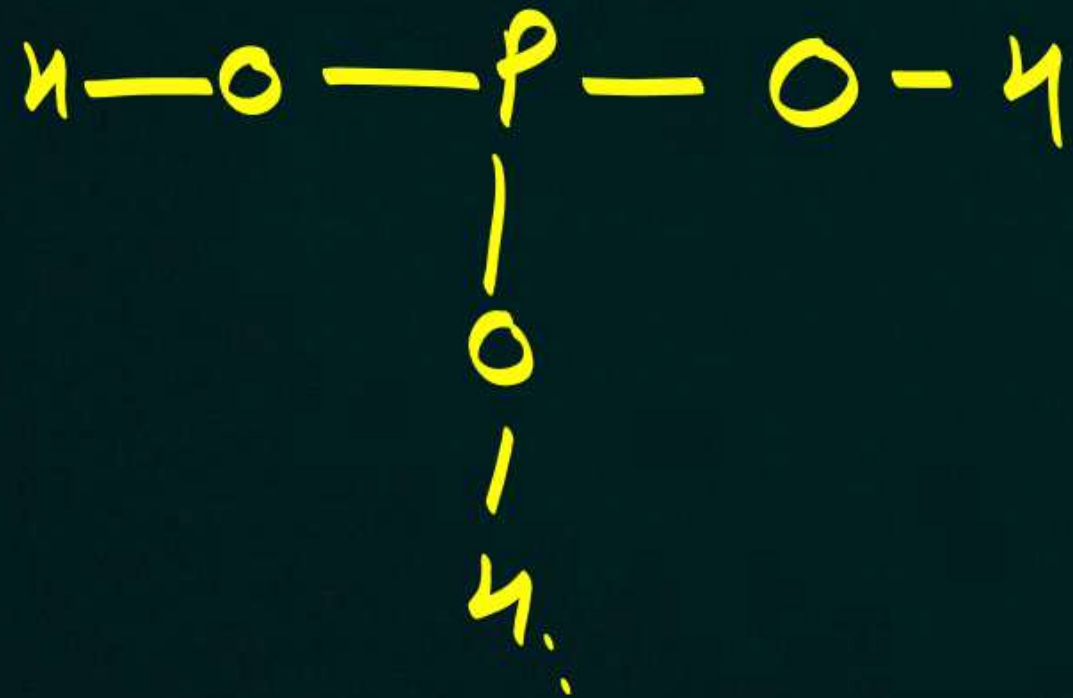
$$\text{Resonance energy} = \text{Energy of most stable Resonating structure} - \text{Energy of Resonance hybrid}$$

Rules of writing Resonating structure:

- Positions of atoms should not change
- Total no. of lone pairs & bond pairs of e^- should remain same.
- Resonating structures must have approximately similar energy.



Are these two structure resonating structure :->



a) Yes

b) No

QUESTION



How many possible resonance forms exist for the nitrate ion, NO_3^- ?

1 One

2 Two

3 Three

4 Four

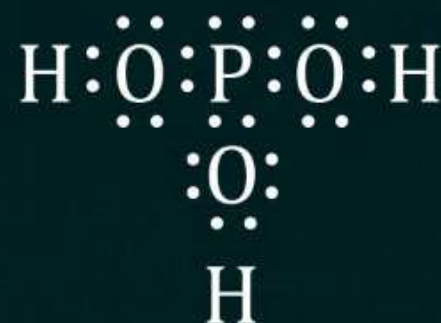
QUESTION



H_3PO_3 can be represented by structures 1 and 2 shown below.



(1)



(2)

However, these two structures cannot be considered as canonical forms of a resonance hybrid because:

- 1** The positions of the atoms are different in each structure.
- 2** The positions of the atoms remain unchanged.
- 3** H_3PO_3 does not show resonance.
- 4** Two hydrogen atoms are missing.

QUESTION

An incorrect statement regarding resonating structures is:

- 1 The contributing structure must have the same number of unpaired electrons. (T)
- 2 The contributing structures should have similar energies. (T)
- 3 The resonance hybrid should have a higher energy than any of the contributing structures. (F)
- 4 None of the above.

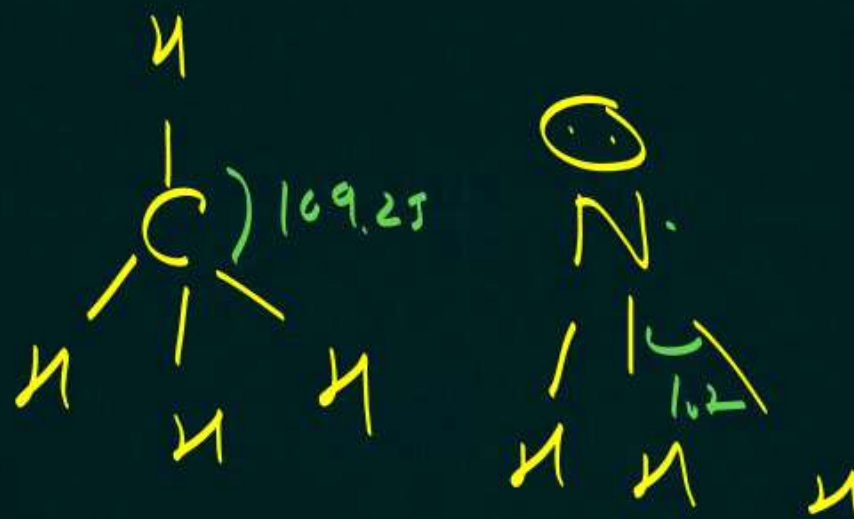
QUESTION

PYQ



The angle between hydrogen through oxygen in H_2O

- 1 104.5°
- 2 104.9°
- 3 105.4°
- 4 105°



QUESTION



Write the resonance structures for SO_3 , NO_2 and NO_3^- .

$\equiv R_1$

$\equiv$

Thank
You